251110

Impedance

Adding Impedances:

$Z_L = j\omega L \qquad Z_C = \frac{1}{j\omega C} \qquad Z_R = R$

$$\boxed{Z_{eq,--} = Z_1 + Z_2 + \dots \quad \text{(series)}}$$

$$\boxed{Z_{eq,\parallel} = \left(\frac{1}{Z_1} + \frac{1}{Z_2} + \dots\right) \quad \text{(Parallel)}}$$

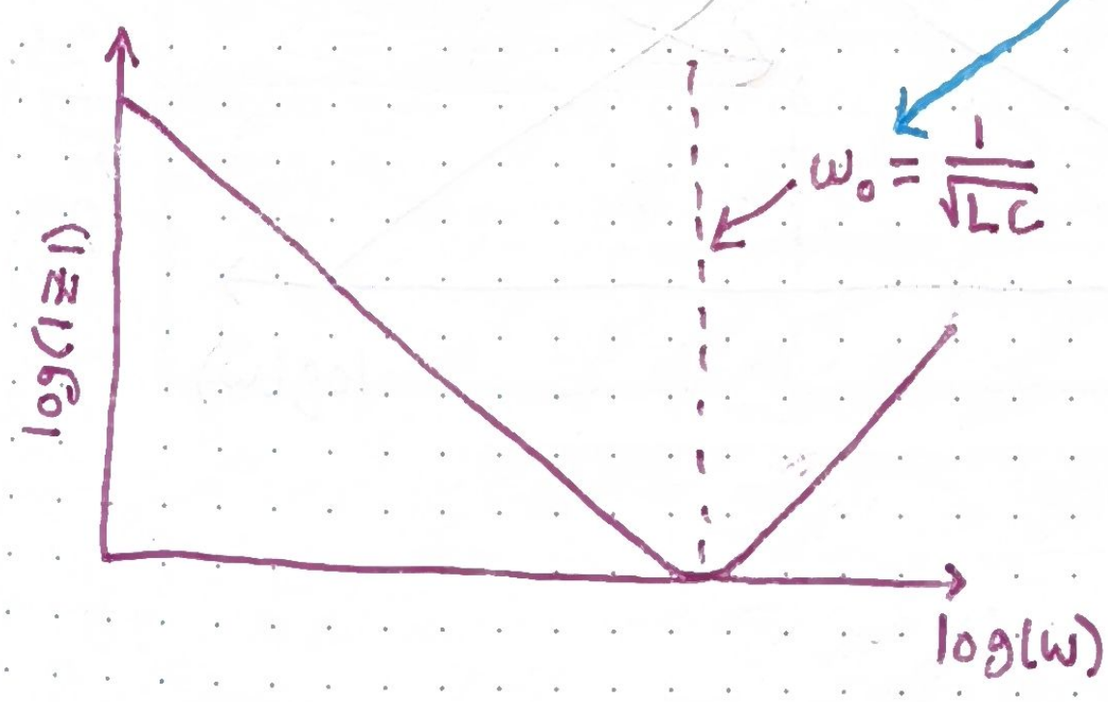
ex./

$$\Rightarrow Z_{ab} = R + \frac{1}{j\omega C} + j\omega L$$

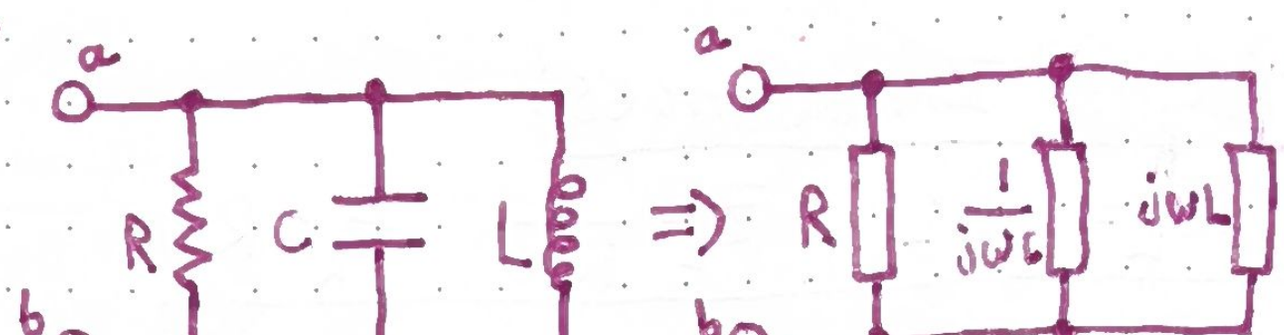
$$= R - j\left(\frac{1}{\omega C}\right) + j\omega L = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$\boxed{\therefore Z_{ab} = \frac{\omega RC + j(\omega^2 LC - 1)}{\omega C}}$$

ω_0 is when Z_{ab} is real



ex./

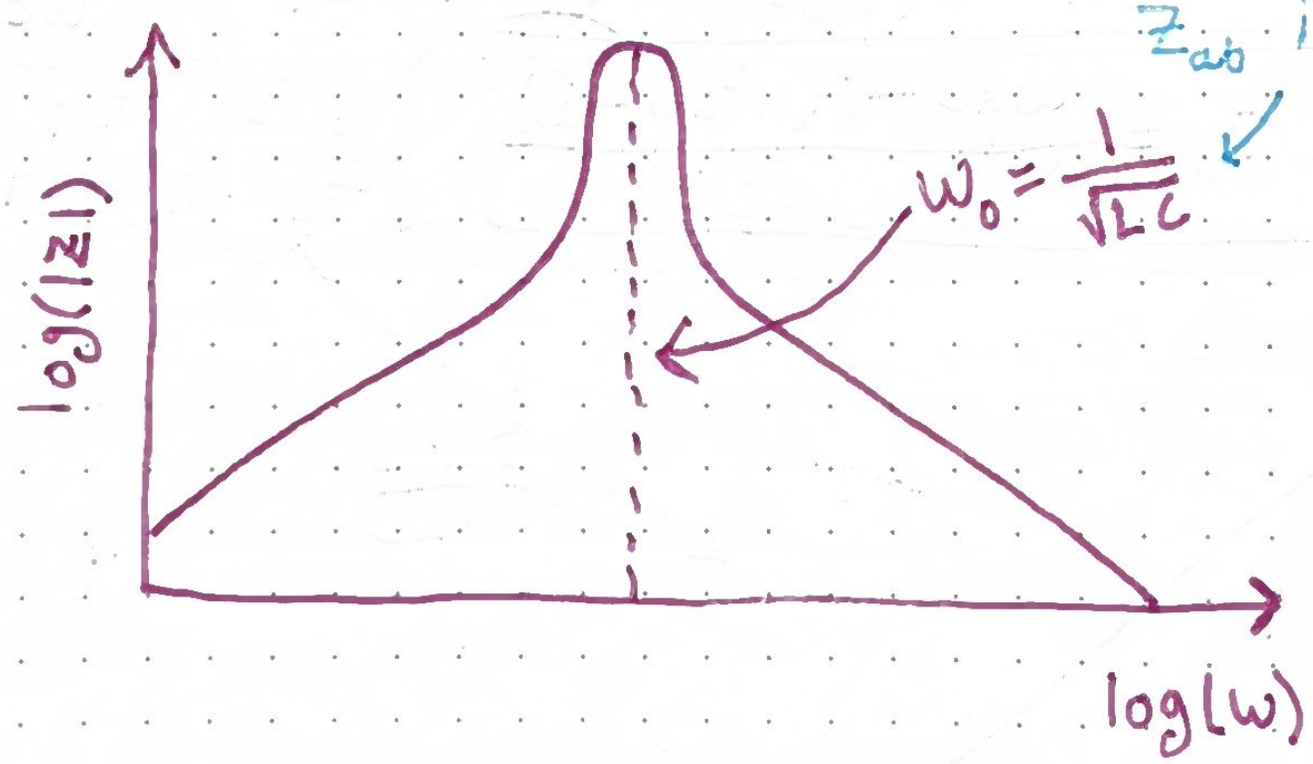


$$\Rightarrow Z_{ab} = \left(\frac{1}{R} + j\omega C + \frac{1}{j\omega L}\right)^{-1} = \left(\frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)\right)^{-1}$$

$$\Rightarrow Z_{ab} = \frac{1}{1/R + j\left(\omega C - \frac{1}{\omega L}\right)}$$

$$\boxed{\therefore\ Z_{ab} = \frac{\omega L}{\omega \frac{L}{R} + j(\omega^2 LC - 1)}}$$

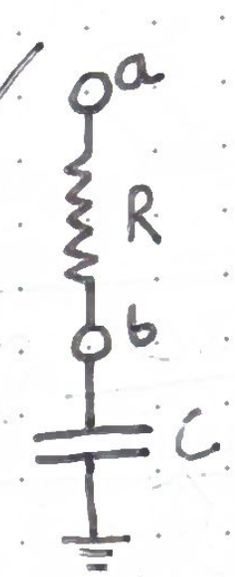
ω_0 is when Z_{ab} is real



Voltage Divis

$$\tilde{V}_z = \frac{Z}{Z_{eq}} \tilde{V}_{eq}$$

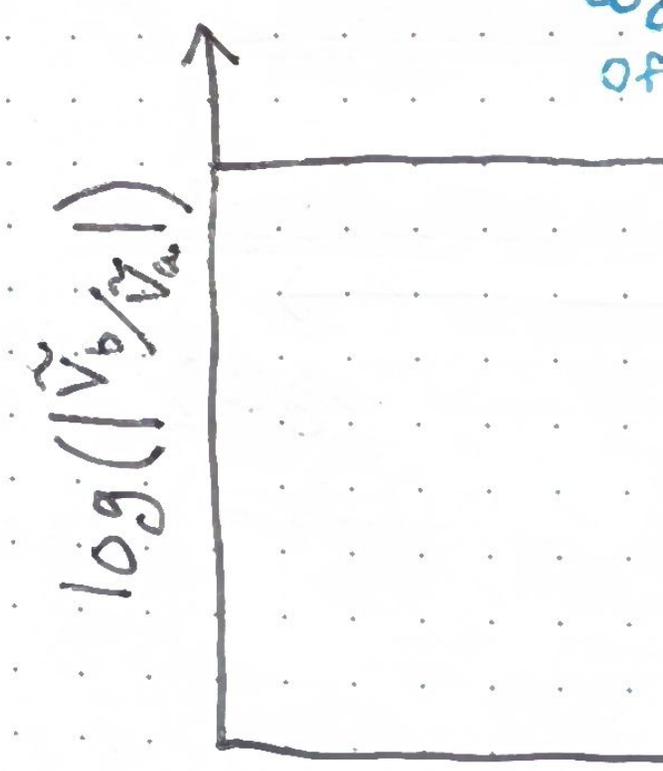
ex./

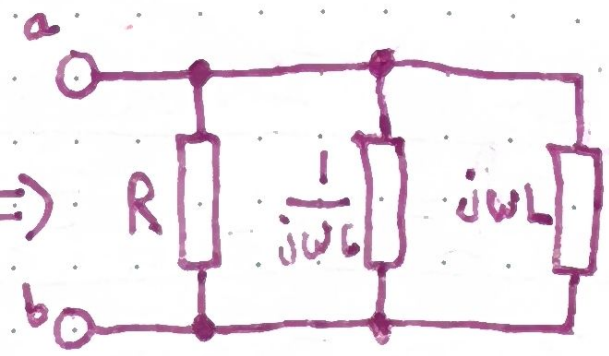


⇒

$$\Rightarrow \tilde{V}_b = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} \tilde{V}$$

ω_0 of





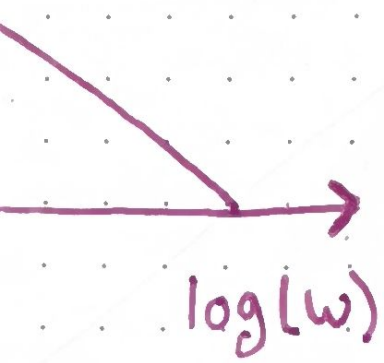
$$\Rightarrow \quad ^{-1} = \left(\frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)\right)^{-1}$$

$$\frac{}{\frac{1}{\omega L})}$$

$$\boxed{\frac{}{LC - 1)}}$$

ω_0 is when Z_{ab} is real

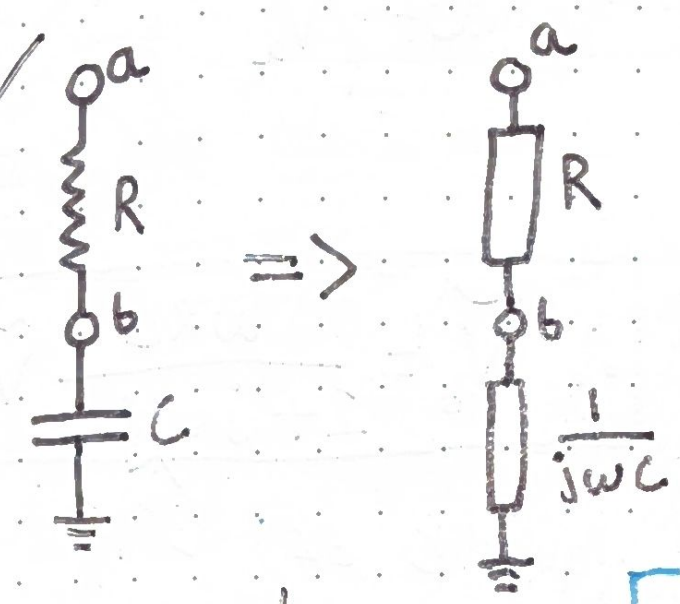
$$\omega_0 = \frac{1}{\sqrt{LC}}$$



Voltage Division:

$$\tilde{V}_z = \frac{Z}{Z_{eq}} \tilde{V}_{eq}$$

ex./

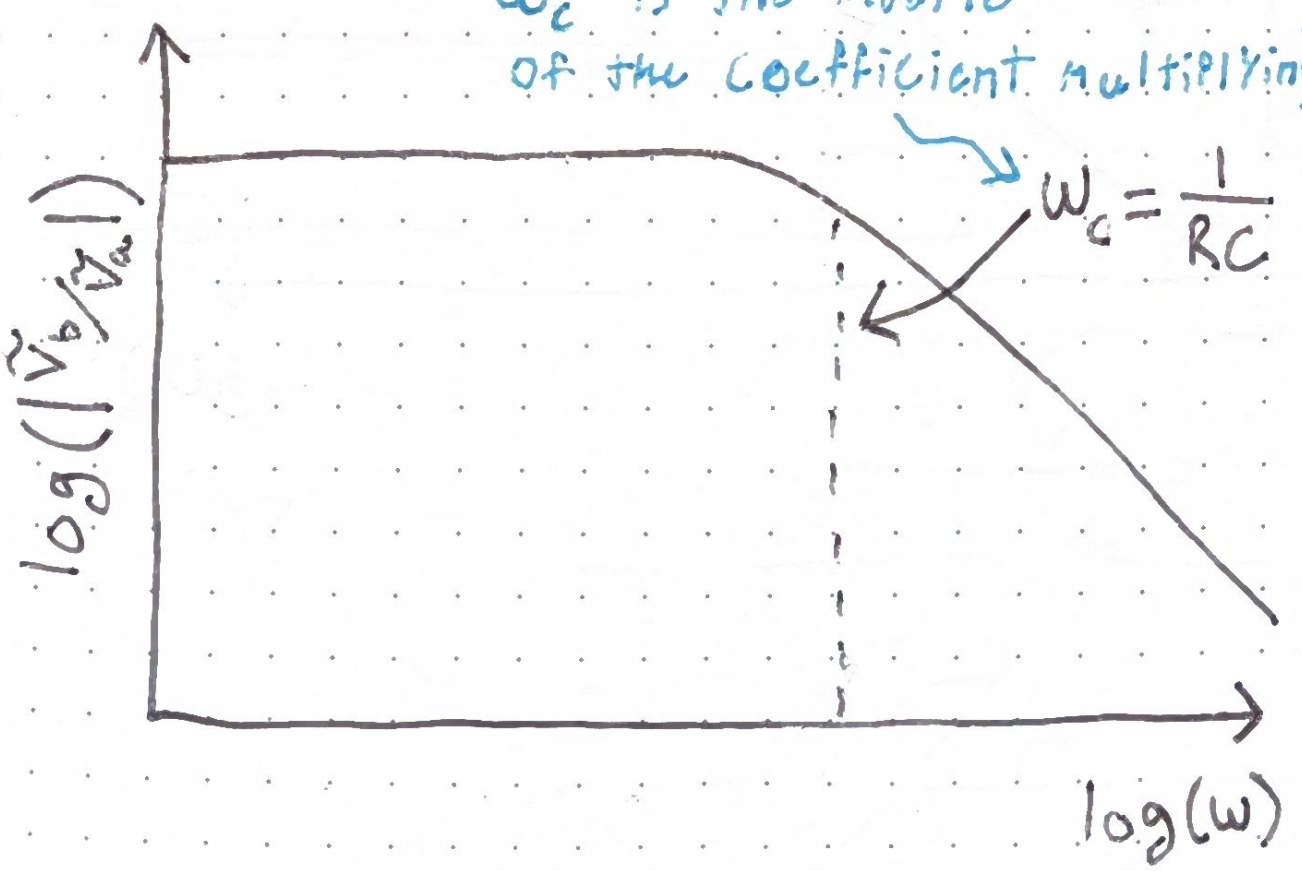


• Find $\tilde{V}_b$ in terms of $\tilde{V}_a$

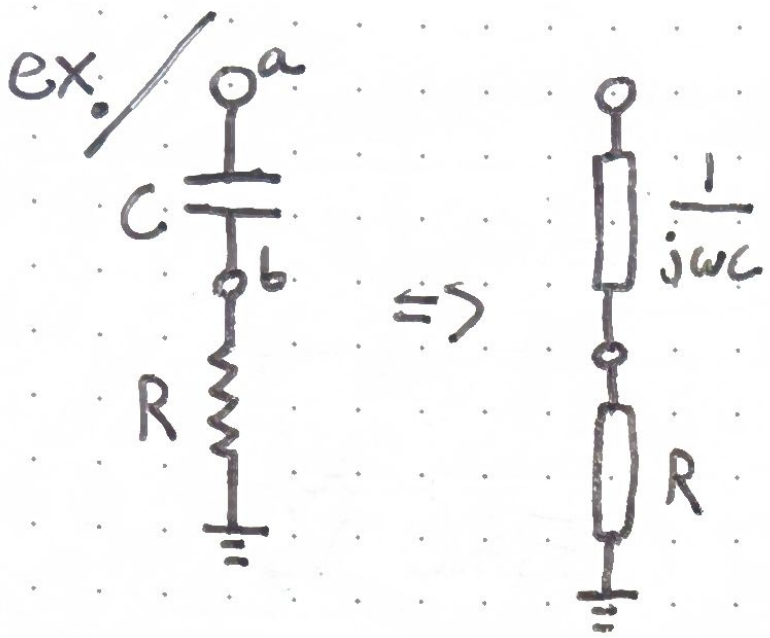
$$\Rightarrow \tilde{V}_b = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} \tilde{V}_a$$

$$\boxed{\therefore \tilde{V}_b = \frac{1}{1 + j\omega RC} \tilde{V}_a}$$

ω_c is the inverse of the coefficient multiplying ω



ex./



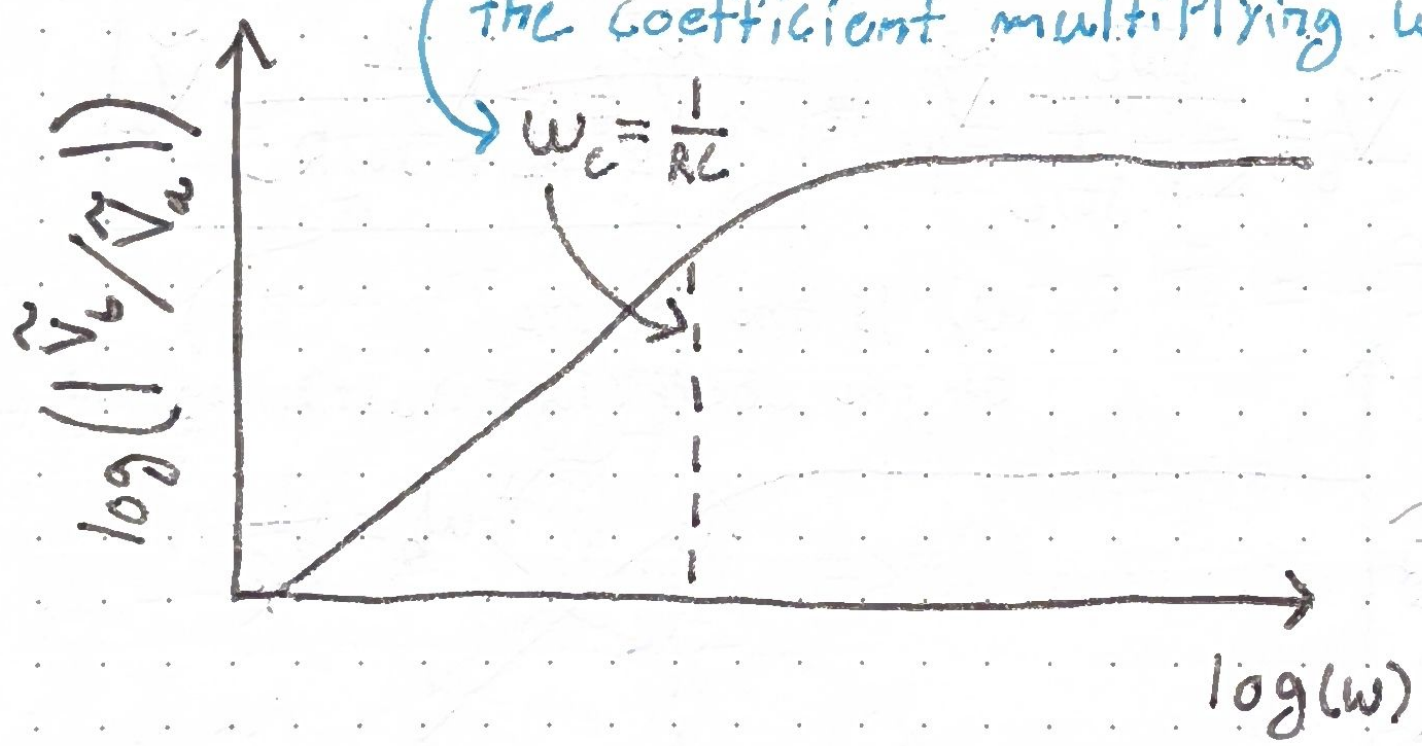
• Find $\tilde{V}_b$ in terms of $\tilde{V}_a$

$$\Rightarrow \tilde{V}_b = \frac{R}{R + \frac{1}{j\omega C}} \tilde{V}_a$$

$$\boxed{\therefore \tilde{V}_b = \frac{\omega RC}{1 + j\omega RC} \tilde{V}_a}$$

ω_c is the inverse of the coefficient multiplying ω

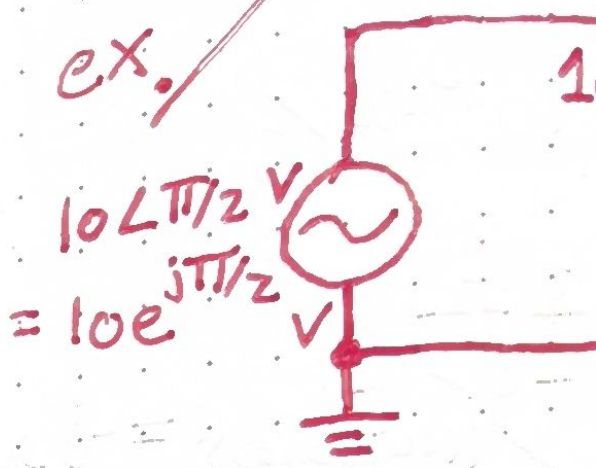
$$\omega_c = \frac{1}{RC}$$



Thévenin Eq

$\tilde{V}_{Th} = \tilde{V}_{oc}$

ex./



$\Rightarrow Z_L = j\omega L$

$\Rightarrow Z_R = 1\,\Omega$

→ Find $\tilde{V}_{oc}$ w/

$\tilde{V}_{oc} = \frac{1}{1+j} 10$

$\Rightarrow \tilde{V}_{oc} = \tilde{V}_{Th} =$

→ Find $\tilde{I}_{sc}$:

→ Find Z_{Th}: Z

$\therefore$

• Find $\tilde{V}_b$ in terms of $\tilde{V}_a$

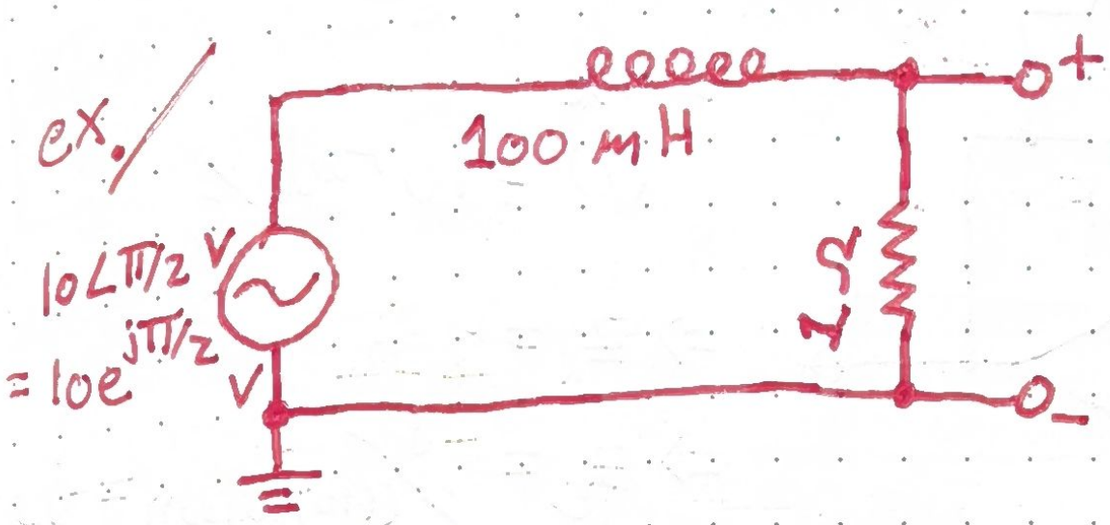
$$\therefore \tilde{V}_b = \frac{\omega RC}{1+j\omega RC}\tilde{V}_a$$

inverse of coefficient multiplying ω

Thévenin Equivalents:

$$\tilde{V}_{Th} = \tilde{V}_{oc} \qquad Z_{Th} = \tilde{V}_{oc}/\tilde{I}_{sc}$$

ex./



and $\omega = 10$ rad/s

$\Rightarrow Z_L = j\omega L = j(10)(0.100) \Rightarrow Z_L = j\Omega$

$\Rightarrow Z_R = 1\Omega$ and $10e^{j\pi/2} = 10j$ V

→ Find $\tilde{V}_{oc}$ w/ volt division:

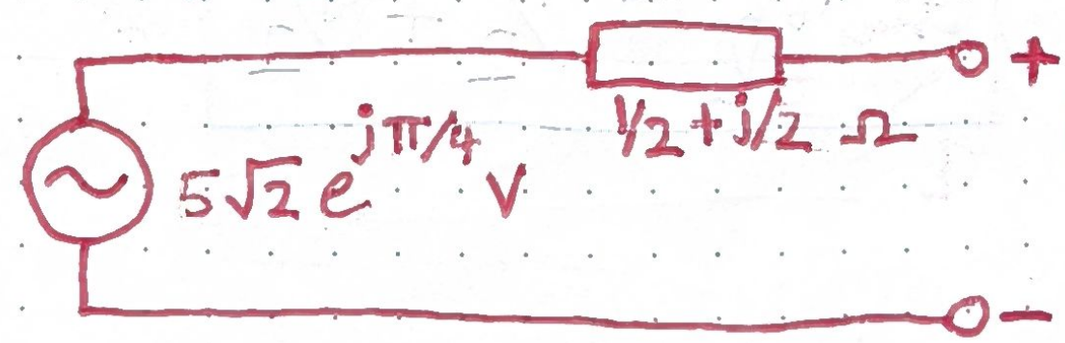
$$\tilde{V}_{oc} = \frac{1}{1+j}10j = \frac{10j}{1+j}\left(\frac{1-j}{1-j}\right) = \frac{10+10j}{2} \text{ V}$$

$\Rightarrow \tilde{V}_{oc} = \tilde{V}_{Th} = 5+5j$ V $\quad \therefore \tilde{V}_{Th} = 5\sqrt{2}e^{j\pi/4}$ V

→ Find $\tilde{I}_{sc}$: $\tilde{I}_{sc} = \frac{10j}{j}$ $\quad \therefore \tilde{I}_{sc} = 10$ A

→ Find Z_{Th}: $Z_{Th} = (5+5j)/10$ $\quad \therefore Z_{Th} = \frac{1}{2} + \frac{j}{2}\ \Omega$

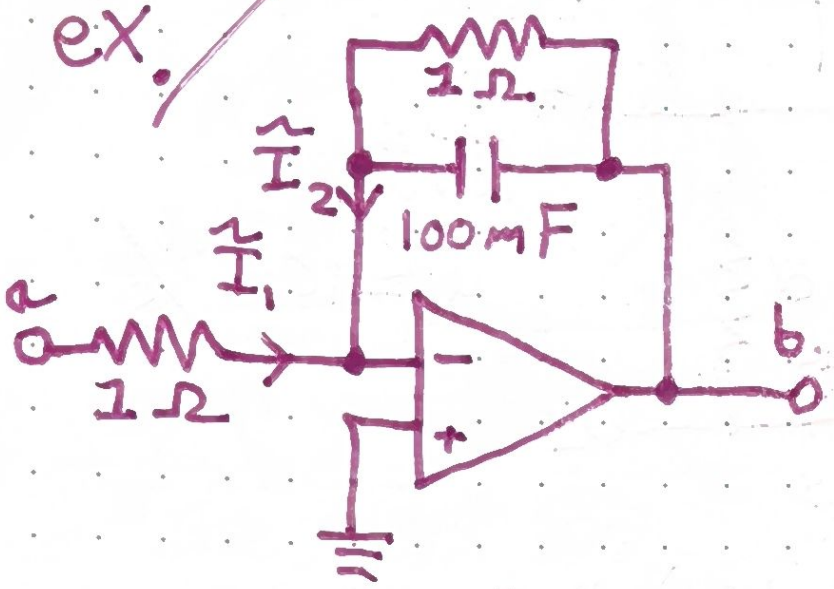
$\therefore$



OP-amps:

$\tilde{V}_p = \tilde{V}_n$ and $\tilde{I}_p = \tilde{I}_n = 0$

ex.



and $\omega = 10\ \text{rad/s}$

$\Rightarrow Z_R = 1\ \Omega$

$\Rightarrow Z_C = \dfrac{-j}{(10)(0.100)} = -j\ \Omega$

$$Z_{R\|C} = \left(\frac{1}{1} + \frac{1}{-j}\right)^{-1} = (1+j)^{-1}$$

$$= \frac{1}{1+j}\left(\frac{1-j}{1-j}\right) \Rightarrow Z_{R\|C} = \frac{1}{2} - \frac{j}{2}\ \Omega$$

→ Use KCL at inverting input: $\tilde{I}_1 = -\tilde{I}_2$

→ Sub. in Ohm's Law:

$$\frac{\tilde{V}_a - 0}{1} = \frac{-(\tilde{V}_b - 0)}{1/2 - j/2}$$

$$\Rightarrow \tilde{V}_a = \frac{-\tilde{V}_b}{1/2 - j/2} \Rightarrow -\tilde{V}_a(1/2 - j/2) = \tilde{V}_b$$

$$\boxed{\therefore\ \tilde{V}_b / \tilde{V}_a = -1/2 + j/2}$$

251112 TM

Impedanc

$$\boxed{Z(\omega) = \frac{\tilde{V}(\omega)}{\tilde{I}(\omega)}}$$

ex. $\tilde{V}_{in}$ ju

$\Rightarrow \dfrac{\tilde{V}_{out}}{\tilde{V}_{in}} = \dfrac{}{j\omega}$

$\Rightarrow \boxed{H(\omega) =}$

*See

Notice:

$|\tilde{V}_{out}| =$

an

$\angle \tilde{V}_{out} =$